1829

Civil engineer Robert Stephenson designed the *Rocket* steam locomotive, which gained worldwide fame.

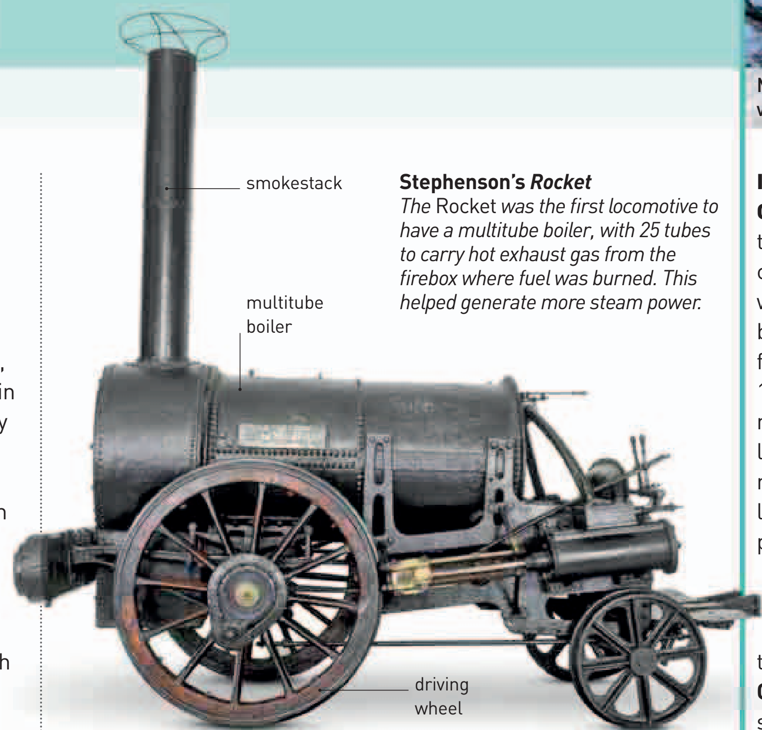
29mph

THE **MAXIMUM SPEED** REACHED BY THE ***ROCKET*** DURING THE **LOCOMOTIVE TRIALS** AT **RAINHILL**, LANCASHIRE, UK

IN NEW YORK, AMERICAN SCIENTIST JOSEPH HENRY (1797–1878) was exploring the power of electromagnetism (see 1820). He found that by carefully insulating the wires and winding them closely and in several layers, he could make very strong electromagnets. In December 1830, Henry finally demonstrated the first powerful electromagnet, able to hold up 750 lb (340 kg) of iron.

In October, the directors of the pioneering Liverpool and Manchester Railway (L&MR) held **locomotive trials at Rainhill** in Lancashire, UK. The public was yet to be convinced of the merits of steam locomotives, so the competition was held to generate publicity as well as to choose which experimental locomotive would pull L&MR's trains. The trials were a huge success. Although only one of the five competing locomotives completed the event—**Robert Stephenson's *Rocket***—the steam age had arrived.

As steam engines developed, engineers became interested in extracting maximum efficiency from them, and theoretical scientists began to **explore the concept of energy.** French scientist **Gustave-Gaspard Coriolis** (1792–1843) published *Calcul de l'Effet des Machines* (*Calculation of the Effect of Machines*), in which he looked at the relationship between energy and work (the effect of energy), and introduced **the concept of kinetic energy**—energy that is produced when an object is in motion (see 1847–48).



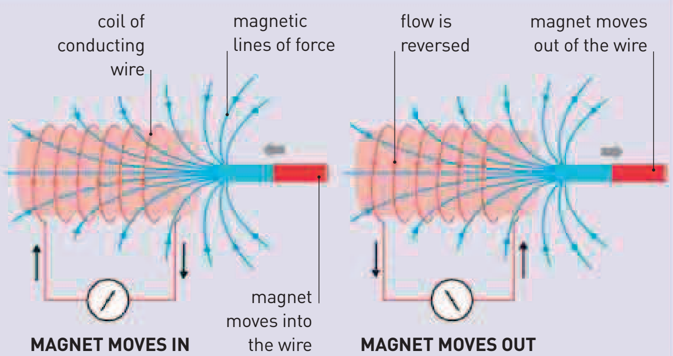
Stephenson's *Rocket*
The Rocket *was the first locomotive to have a multitube boiler, with 25 tubes to carry hot exhaust gas from the firebox where fuel was burned. This helped generate more steam power.*

The **identification of geological periods** gathered pace. While studying sediments in the Seine valley in France, geologist **Jules Desnoyers** (1800–87) coined **the term Quaternary** to describe the most recent geological period, when loose material (gravel, sand, and clay deposits) was laid down on top of solid rocks.

In a cave in Engis, Belgium, Dutch–Belgian prehistorian **Philippe-Charles Schmerling** (1791–1836) found a fragment of a small child's skull. This was only the **second discovery of a human fossil**; British geologist William Buckland had discovered the first in 1823. Schmerling's find later proved to be 30,000 to 70,000 years old—the **first Neanderthal remains ever discovered**.

ELECTROMAGNETIC INDUCTION

This demonstration of electromagnetic induction involves moving a bar magnet in and out of a coil of wire to produce an electric current. The magnet's field pulls the electrons in the wire and produces a voltage. If the coil is connected to a circuit, an electric current will flow. If the magnet is moved in the other direction, the flow is reversed.



1830

Metamorphic rocks, such as these, were first identified by Charles Lyell.

IN THE EARLY 19TH CENTURY, GEOLOGISTS WERE DIVIDED into two groups. The **catastrophists** claimed that the surface of Earth was shaped by a few huge and brief catastrophes, such as floods and earthquakes (see 1812). The **uniformitarians**, meanwhile, believed that Earth's landscapes were shaped and reshaped gradually over very long periods of time by steady processes, such as river erosion.

More evidence was being found confirming the uniformitarian school of thought. Scottish geologist **Charles Lyell** (1797–1875) summarized these findings, insisting that change was continuous and gradual, in his monumental book *Principles of Geology,* which was published in three parts between 1830 and 1833. Lyell's text was so authoritative and convincing that after its publication, few doubted that the **Earth had gone through many geological ages** over millions of years. It was this picture of Earth's vast geological history that paved the way for Charles Darwin's theory of evolution (see 1859), which was partly influenced by Lyell's work.

The same year, German astronomer **Johann von Mädler** (1794–1874) began to make **drawings of the surface of Mars**. These were later

Jules Desnoyers coins the term **Quaternary**

Gustave-Gaspard Coriolis coins the term **kinetic energy**

Philippe-Charles Schmerling discovers a **fossilized Neanderthal skull fragment**

July 23 American inventor William Burt patents a **typewriter**

Charles Lyell publishes the first volume of ***Principles of Geology***

German physician Johann Lukas Schönlein coins the term **hemophilia**

October Robert Stephenson's *Rocket* wins the Rainhill locomotive trials

Johann von Mädler and Wilhelm Beer make the **first map of Mars**

Giovanni Amici confirms the function of the **pollen tube**